

Project Report Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	13th july 2026
Team ID	+SWTID-2026-1292
Project Name	Heart Disease analysis
Maximum Marks	5 Marks

HEART DISEASE ANALYSIS USING TABLEAU

Project Report

1. INTRODUCTION

1.1 Project Overview

Heart disease is one of the leading causes of death worldwide. Healthcare organizations collect enormous amounts of patient data, but extracting meaningful insights from this data is often difficult using traditional reporting methods. This project, Heart Disease Analysis using Tableau, focuses

on analyzing patient health records and presenting them through interactive dashboards. The dashboards help healthcare professionals understand disease trends, identify risk factors, and support faster clinical decision-making.

The project uses Tableau to transform raw healthcare data into meaningful visualizations such as bar charts, line charts, pie charts, scatter plots, and KPI cards. These visualizations enable users to explore relationships between various health parameters including age, cholesterol, blood pressure, chest pain type, and heart disease diagnosis.

1.2 Purpose

The main purpose of this project is to develop an interactive business intelligence dashboard that helps doctors, researchers, and hospital administrators analyze heart disease data efficiently. The dashboard provides clear insights into disease prevalence, patient demographics, and medical risk factors, enabling data-driven healthcare decisions.

2. IDEATION PHASE

2.1 Problem Statement

Healthcare institutions generate vast amounts of patient information every day. Analyzing this information manually is time-consuming, prone to errors, and makes it difficult to identify hidden patterns associated with heart disease.

This project aims to solve this problem by creating an interactive Tableau dashboard that visualizes patient data, identifies high-risk groups, and supports effective clinical decision-making.

[Define Problem Statements Template.docx](#)

2.2 Empathy Map Canvas

Users

- Doctors
- Cardiologists
- Hospital Administrators
- Healthcare Researchers

Says

- I need accurate patient reports.
- I want quick access to patient insights.

Thinks

- Which patients are at high risk?

- How can disease trends be identified quickly?

Does

- Reviews medical reports
- Examines patient history
- Uses Tableau dashboards

Feels

- Concerned about patient health
- Wants faster diagnosis
- Needs reliable analytics

Pain Points

- Manual data analysis
- Large healthcare datasets
- Difficult trend identification

Gains

- Interactive dashboards
- Better diagnosis
- Improved patient care
- Faster decision-making

2.3 Brainstorming

Several ideas were considered during the ideation phase:

- Age-wise heart disease analysis
- Gender-based disease comparison
- Cholesterol level analysis
- Blood pressure monitoring
- Chest pain analysis
- Interactive Tableau dashboards
- KPI performance monitoring

The final idea selected was developing an interactive Tableau dashboard for comprehensive heart disease analysis

[BrainstormingTemplate](#) .

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Awareness

Healthcare professionals identify the need for better patient data analysis.

Data Collection

Patient records are collected from hospitals or publicly available datasets.

Data Cleaning

Missing values and duplicate records are removed.

Dashboard Creation

Interactive dashboards are developed using Tableau.

Decision Making

Healthcare professionals use dashboard insights for diagnosis and treatment planning.

[Customer Journey Map - Example](#)

3.2 Solution Requirement

Functional Requirements

- Import patient dataset
- Clean and preprocess data
- Create interactive dashboards

- Apply filters
- Generate reports

Non-functional Requirements

- Easy to use
- Fast dashboard loading
- Accurate visualization
- Reliable performance
- Scalability

3.3 Data Flow Diagram

Patient Dataset

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Data Cleaning

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Data Transformation

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Tableau Dashboard

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Charts & KPIs

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Healthcare Insights

3.4 Technology Stack

Component	Technology
Visualization	Tableau Desktop
Dataset	CSV / Excel
Data Preparation	Tableau Prep / Excel
Storage	Local File System
Charts	Tableau

4. PROJECT DESIGN

4.1 Problem–Solution Fit

Problems

- Large healthcare datasets
- Manual analysis
- Difficult trend identification
- Lack of interactive reports

Solutions

- Data preprocessing
- Tableau dashboards
- KPI visualization
- Interactive filters

Benefits

- Better patient care
- Faster diagnosis
- Improved decision-making
- Interactive reporting

4.2 Proposed Solution

The proposed solution uses Tableau Desktop to analyze heart disease data. Patient records are cleaned, transformed, and visualized using interactive dashboards. Users can filter information based on age, gender, cholesterol, chest pain type, and diagnosis to obtain meaningful insights

4.3 Solution Architecture

CSV Dataset



Data Cleaning



Tableau Data Source



Calculated Fields



Dashboards





Reports & Insights

5. PROJECT PLANNING & SCHEDULING

Sprint Plan

Sprint

Task

Sprint 1 Dataset Collection

Sprint 2 Data Cleaning

Sprint 3 Dashboard Development

Sprint 4 Testing & Documentation

6. FUNCTIONAL AND PERFORMANCE TESTING

Data Rendered

Dataset loaded successfully.

Data Preprocessing

- Missing values checked
- Duplicate records removed
- Data transformed

Dashboard Testing

- Filters working correctly
- Charts displaying correctly
- KPI cards validated

Performance

Dashboard loads quickly and supports interactive filtering without significant delay.

7. RESULTS

The Tableau dashboard successfully visualizes heart disease information using:

- Age Distribution
- Gender Analysis
- Cholesterol Analysis
- Blood Pressure Analysis
- Chest Pain Analysis
- Heart Disease Prediction
- KPI Cards
- Interactive Filters

The dashboard enables healthcare professionals to identify patterns and make informed clinical decisions.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Easy to understand
- Interactive dashboards
- Fast data analysis
- Better healthcare insights
- Supports decision making
- Attractive visualizations

Disadvantages

- Depends on data quality
- Requires Tableau software
- Limited predictive capability without machine learning
- Manual data updates for offline datasets

9. CONCLUSION

The Heart Disease Analysis using Tableau project demonstrates how business intelligence tools can transform healthcare data into meaningful insights. Interactive dashboards enable doctors and healthcare administrators to analyze disease trends, identify risk factors, and improve patient care. Tableau provides an efficient platform for visual analytics, supporting faster and more informed decision-making.

10. FUTURE SCOPE

Future enhancements include:

- Machine Learning-based disease prediction
- Real-time hospital database integration
- Cloud deployment using Tableau Server
- Mobile dashboard support
- AI-assisted healthcare recommendations
- Integration with Electronic Health Record (EHR) systems

11. APPENDIX

Dataset

Heart Disease Dataset (UCI Machine Learning Repository / Kaggle)

Software Used

- Tableau Desktop

12.LINKS :

https://public.tableau.com/views/HeartDiseaseAnalysis_TableauPublic/Sheet2?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

13.GITHUB LINK : https://github.com/saiganeshsaiganesh344-sys/Heart-Disease-Analysis_-_Tableau